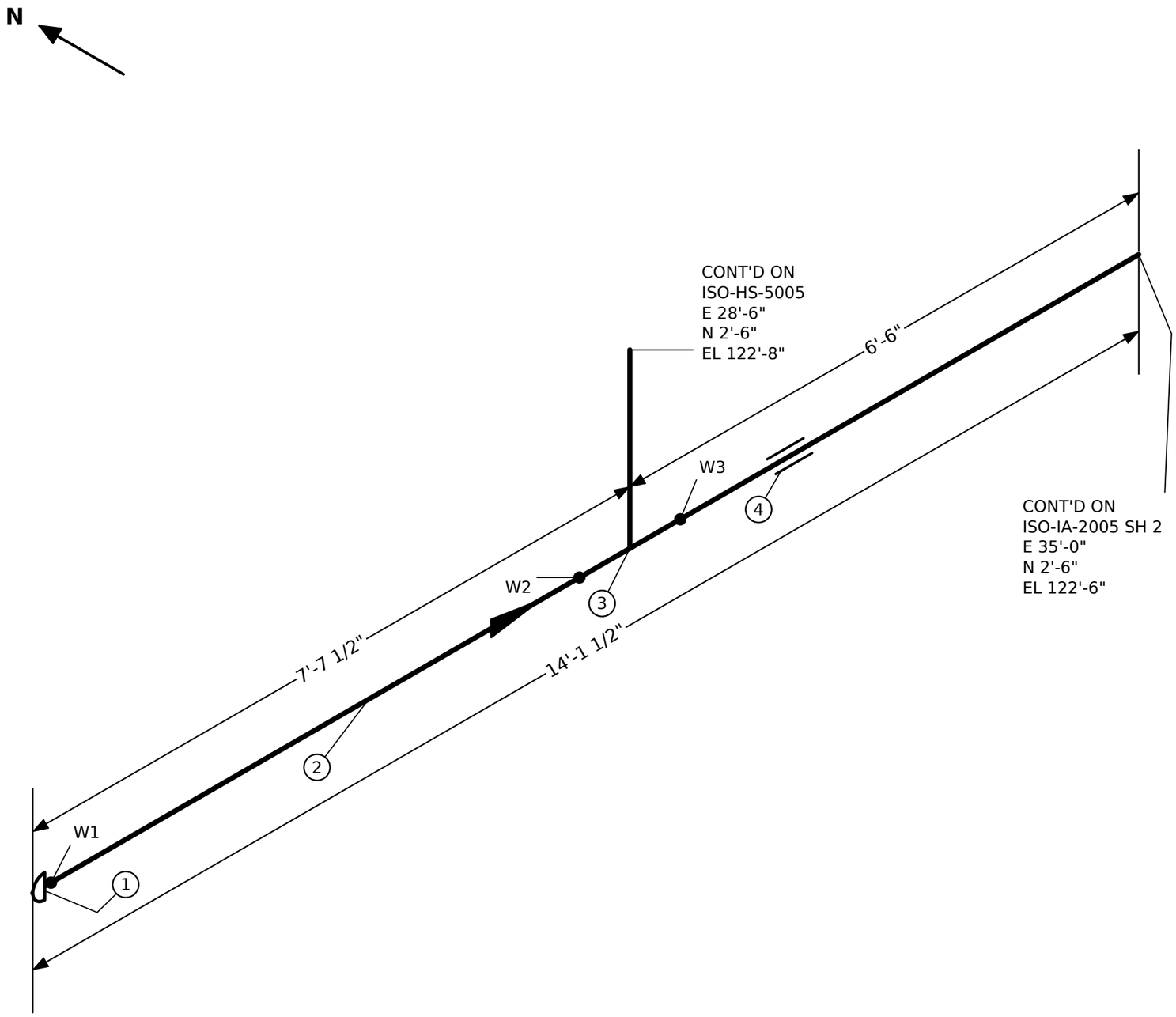


BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	1	2"	CAP 2" P150
2	13'-8"	2"	PIPE 2" P150 BEVELLED ENDS
3	1	2"x1/2"	TEE REDUCING 2" × 1/2" P150
4	1	2"	SHOE FOR 2" LINE (SHOE-2)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

AIR AND GASES

NOTES:

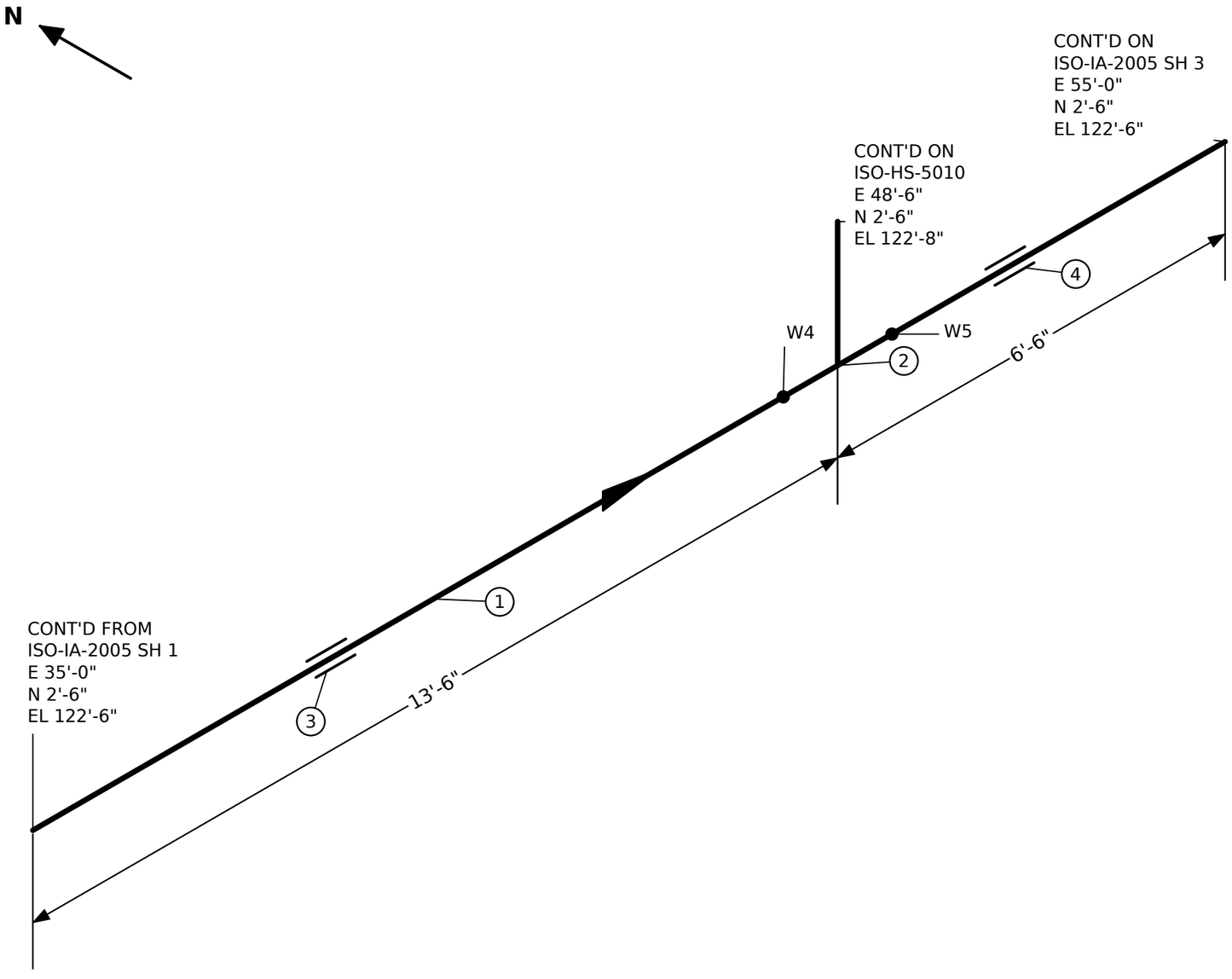
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. PAINT PER PROJECT SPECIFICATION

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	IA

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P150-IA-2005	G150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-IA-2005	1 OF 8	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	19'-8"	2"	PIPE 2" P150 BEVELLED ENDS
2	1	2"x1/2"	TEE REDUCING 2" × 1/2" P150
3	1	2"	GUIDE FOR 2" LINE (GUIDE-2)
4	1	2"	SHOE FOR 2" LINE (SHOE-2)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. PAINT PER PROJECT SPECIFICATION
4. REMOVE BURRS FROM ALL CUT ENDS

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	IA

COMPANY

SMARTWELD

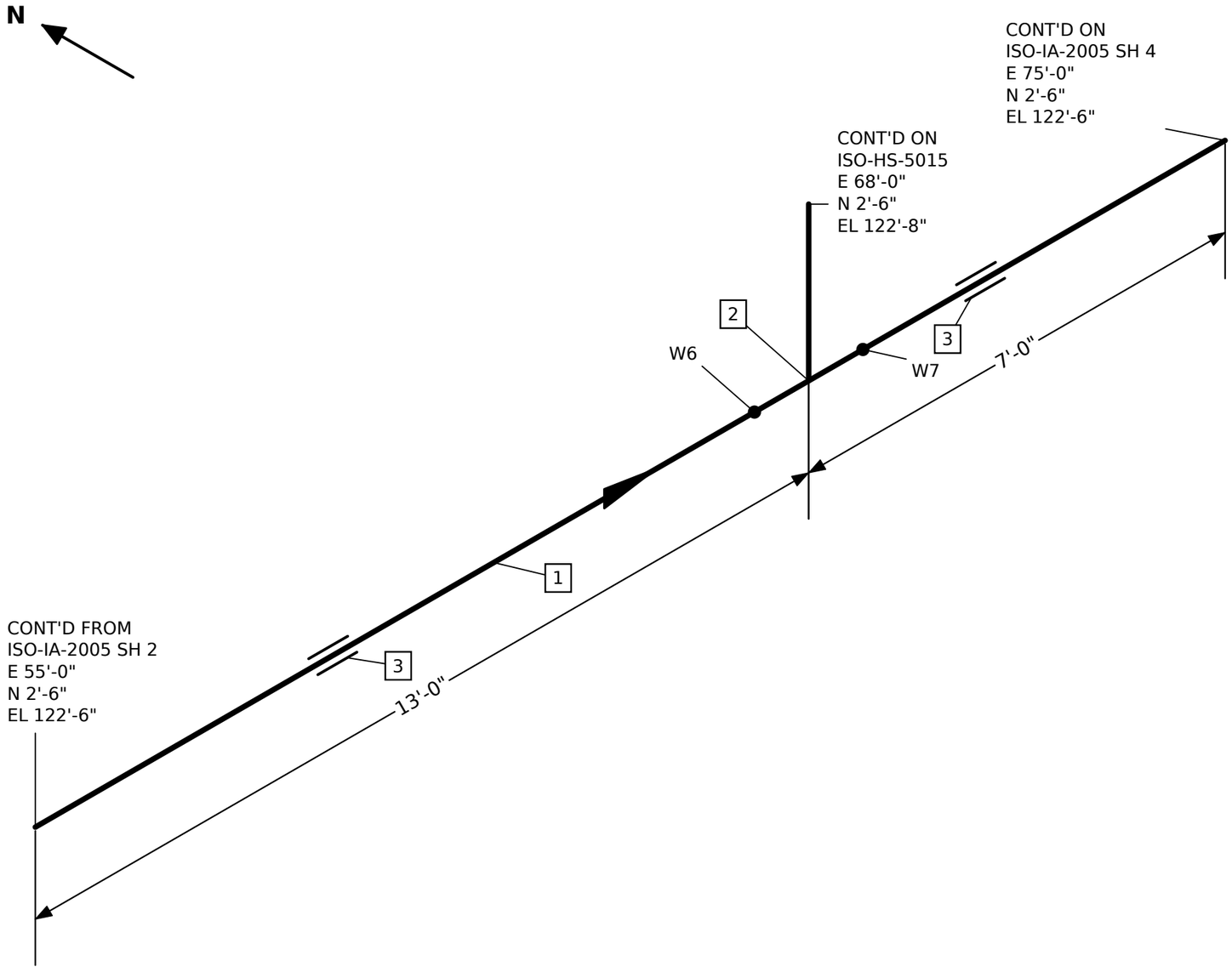
TITLE

AIR AND GASES

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P150-IA-2005	G150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-IA-2005	2 OF 8	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	19'-8"	2"	PIPE 2" P150 BEVELLED ENDS
2	1	2"x1/2"	TEE REDUCING 2" × 1/2" P150
3	2	2"	SHOE FOR 2" LINE (SHOE-2)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. REMOVE BURRS FROM ALL CUT ENDS

TEST PRESS. **-**	INSULATION **NONE**	PAINT **-**	
PWHT **-**	NDE **-**	FLUID **IA**	

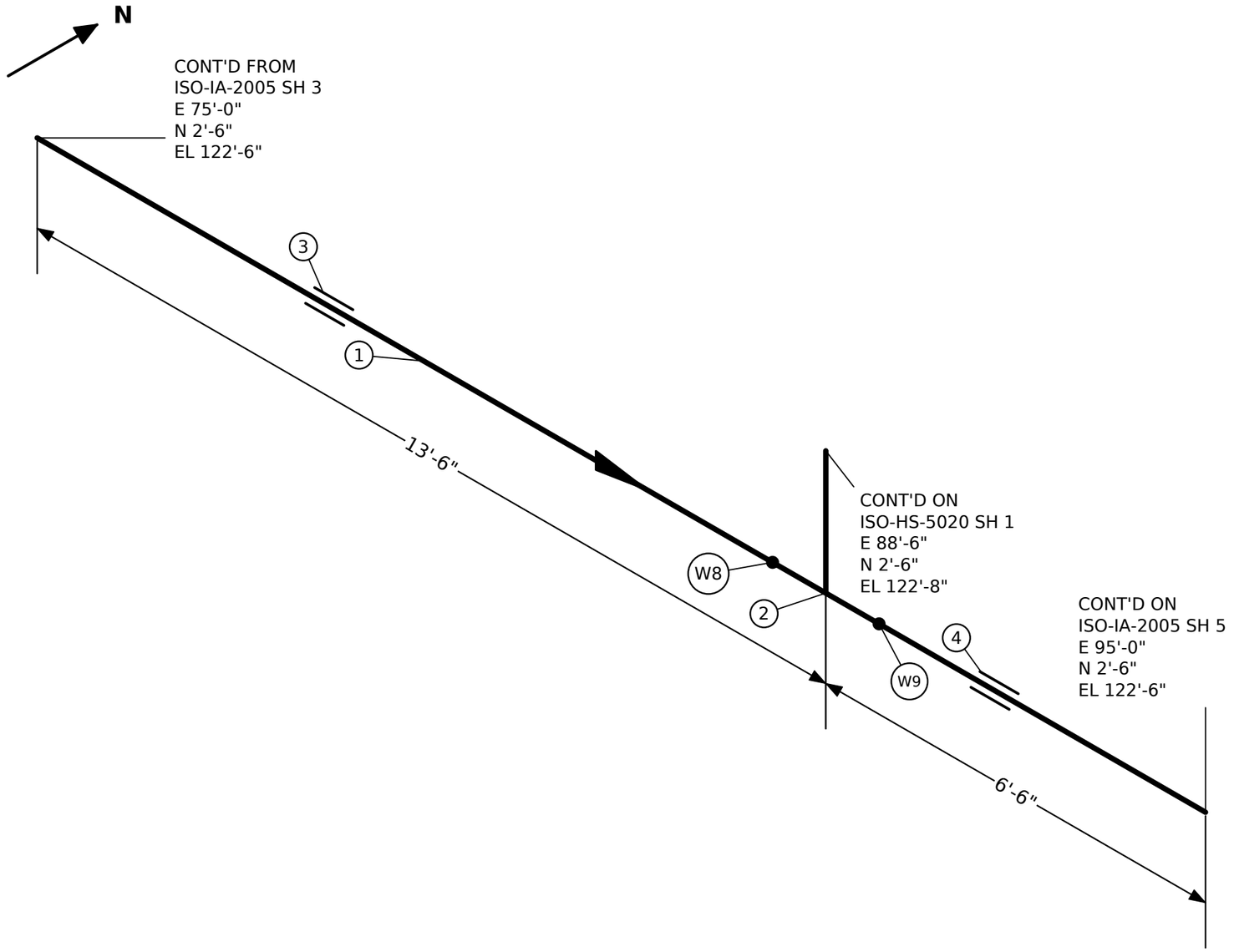
COMPANY

SMARTWELD

TITLE

AIR AND GASES

LINE NO. **2"-P150-IA-2005**	PIPING CLASS **G150**	DESIGN PRESS. **150 PSIG**	DESIGN TEMP. **-**
DRAWING NO. **ISO-IA-2005**		SHEET **3 OF 8**	REV **1**
DATE **2026-11-01**	DRAWN **-**	CHECKED **-**	SCALE **NTS**



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	19'-8"	2"	PIPE 2" P150 BEVELLED ENDS
2	1	2"x1/2"	TEE REDUCING 2" × 1/2" P150
3	1	2"	GUIDE FOR 2" LINE (GUIDE-2)
4	1	2"	SHOE FOR 2" LINE (SHOE-2)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	IA

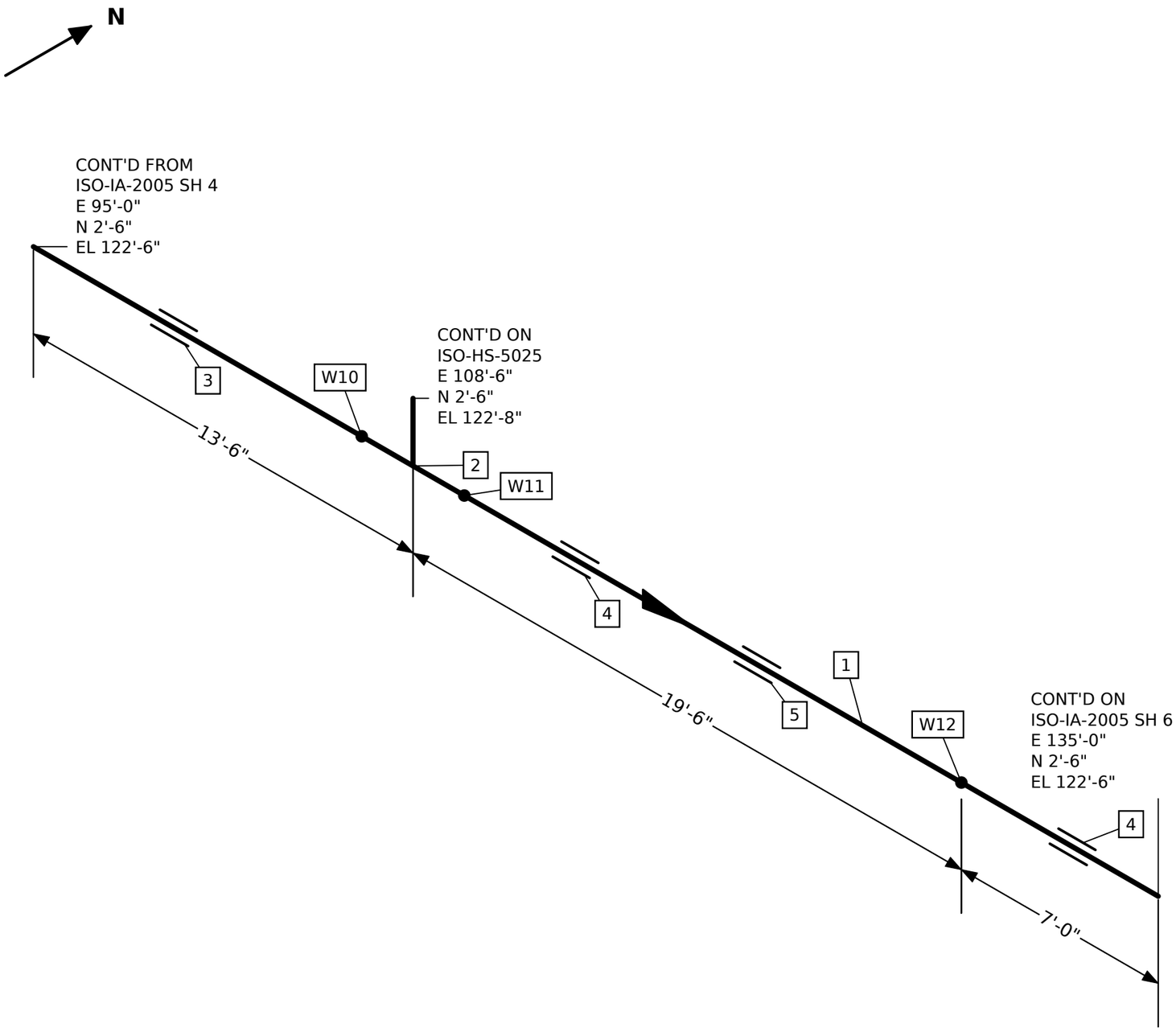
COMPANY
SMARTWELD

TITLE
AIR AND GASES

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P150-IA-2005	G150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-IA-2005	4 OF 8	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	39'-8"	2"	PIPE 2" P150 BEVELLED ENDS
2	1	2"x1/2"	TEE REDUCING 2" × 1/2" P150
3	1	2"	ANCHOR FOR 2" LINE (ANCHOR-2)
4	2	2"	SHOE FOR 2" LINE (SHOE-2)
5	1	2"	GUIDE FOR 2" LINE (GUIDE-2)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. REMOVE BURRS FROM ALL CUT ENDS
4. HYDROTEST BEFORE PAINTING

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	IA

COMPANY

SMARTWELD

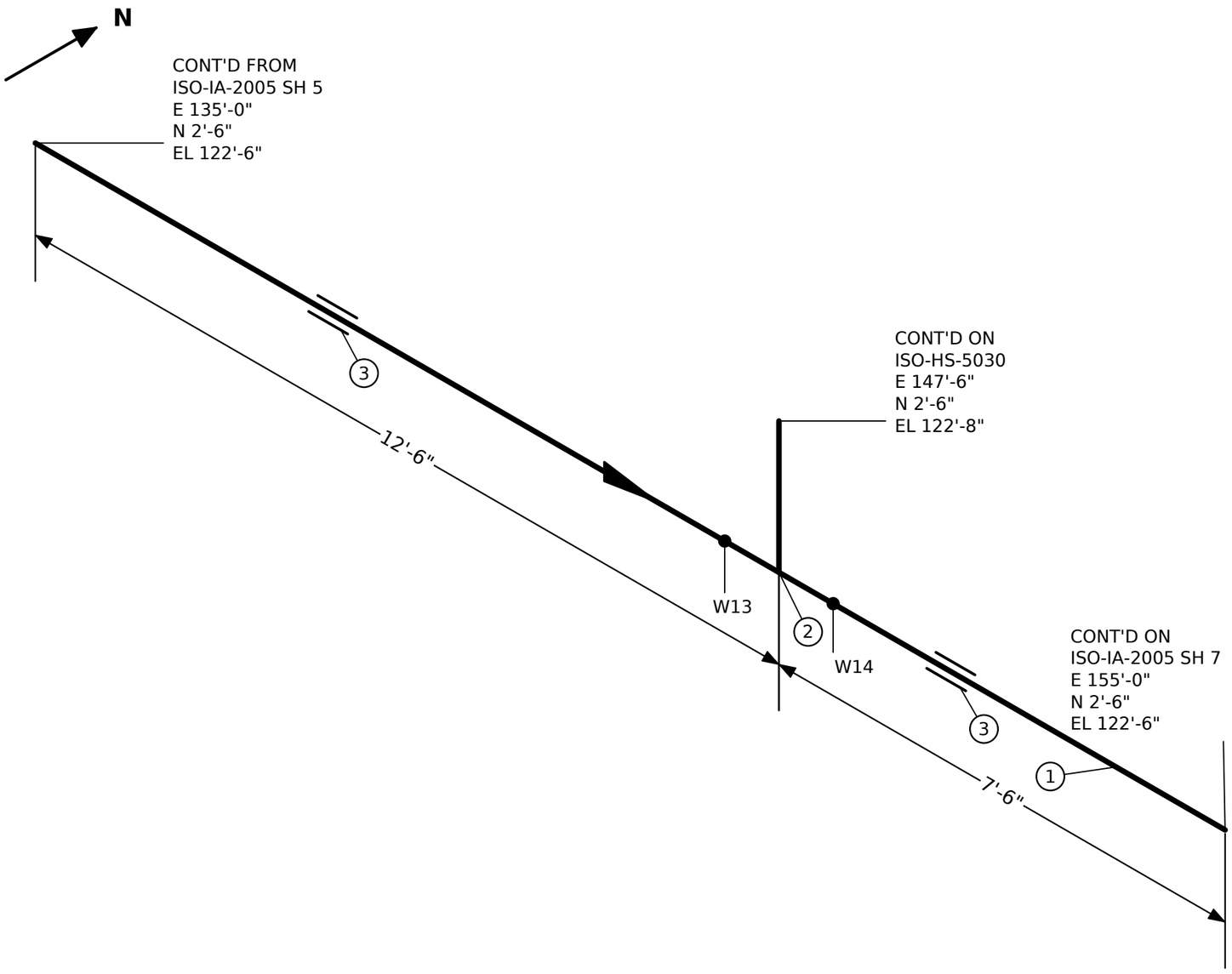
TITLE

AIR AND GASES

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P150-IA-2005	G150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-IA-2005	5 OF 8	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	19'-8"	2"	PIPE 2" P150 BEVELLED ENDS
2	1	2"x1/2"	TEE REDUCING 2" × 1/2" P150
3	2	2"	SHOE FOR 2" LINE (SHOE-2)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

AIR AND GASES

NOTES:

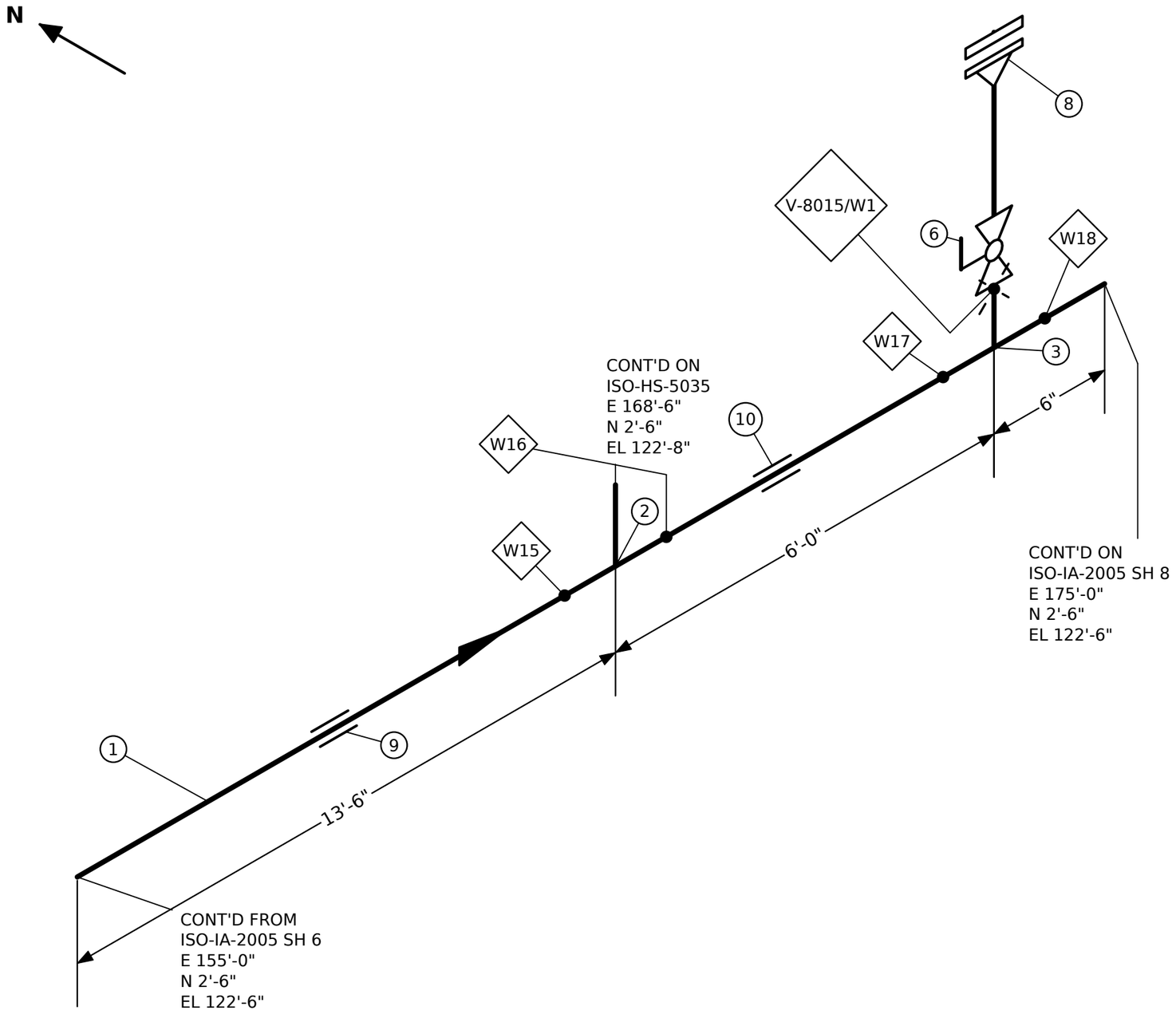
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	IA

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P150-IA-2005	G150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-IA-2005	6 OF 8	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	19'-4"	2"	PIPE 2" P150 BEVELLED ENDS
2	1	2"x1/2"	TEE REDUCING 2" × 1/2" P150
3	1	2"x3/4"	TEE REDUCING 2" × 3/4" P150
4	1	3/4"	FLANGE WELD-NECK 3/4" P150 RAISED FACE
5	2	3/4"	GASKET 3/4" P150 COMPRESSED SHEET
6	1	3/4"	VALVE, BALL, FLOATING 3/4" P150
7	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150
8	1	3/4"	FLANGE BLIND 3/4" P150 RAISED FACE
9	1	2"	GUIDE FOR 2" LINE (GUIDE-2)
10	1	2"	SHOE FOR 2" LINE (SHOE-2)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

AIR AND GASES

NOTES:

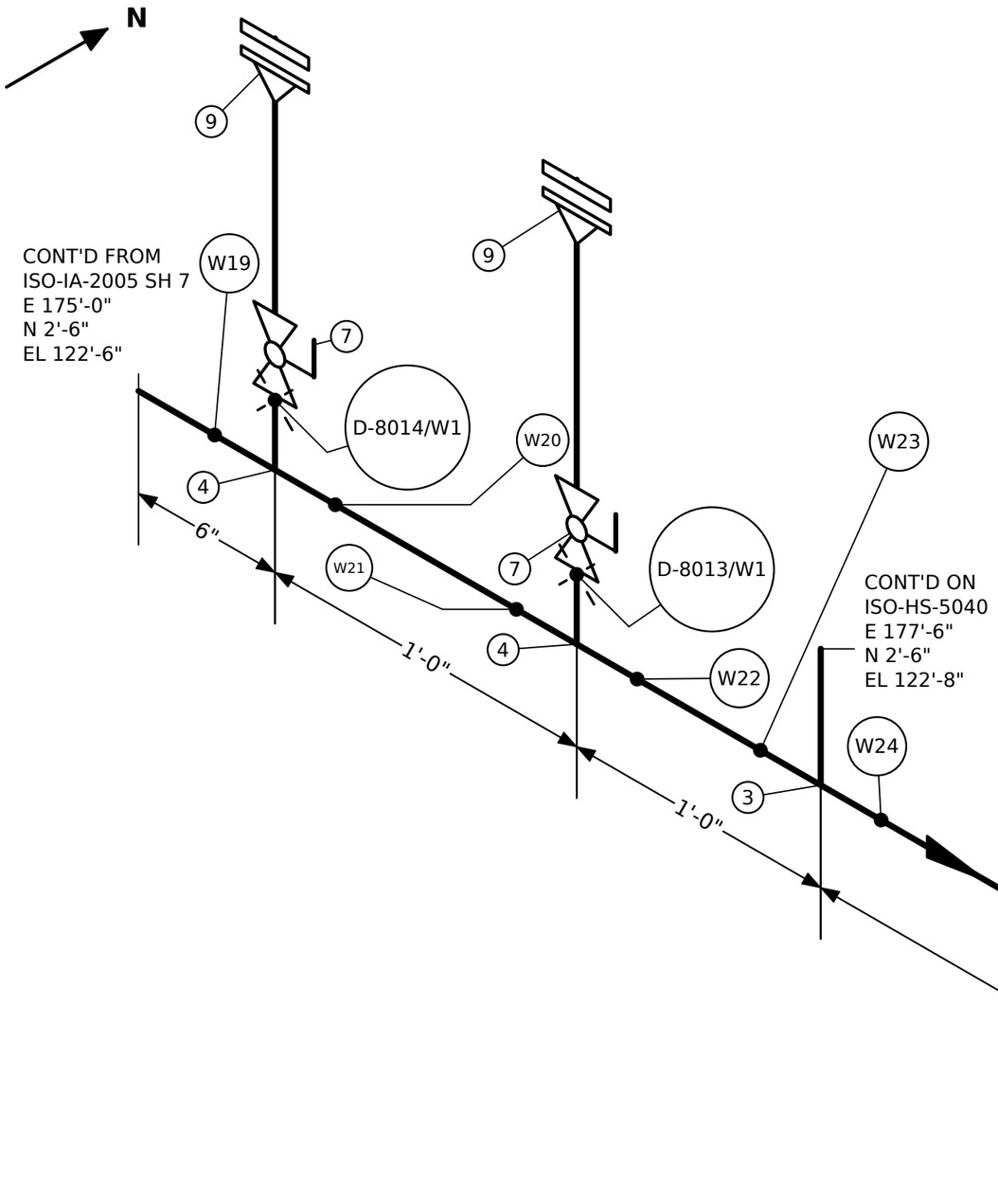
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. REMOVE BURRS FROM ALL CUT ENDS

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	IA

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P150-IA-2005	G150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-IA-2005	7 OF 8	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	1	2"	CAP 2" P150
2	3'-0"	2"	PIPE 2" P150 BEVELLED ENDS
3	1	2"x1/2"	TEE REDUCING 2" × 1/2" P150
4	2	2"x3/4"	TEE REDUCING 2" × 3/4" P150
5	2	3/4"	FLANGE WELD-NECK 3/4" P150 RAISED FACE
6	4	3/4"	GASKET 3/4" P150 COMPRESSED SHEET
7	2	3/4"	VALVE, BALL, FLOATING 3/4" P150
8	4	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150
9	2	3/4"	FLANGE BLIND 3/4" P150 RAISED FACE

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

TEST PRESS. -	INSULATION NONE	PAINT -	
PWHT -	NDE -	FLUID IA	

COMPANY

SMARTWELD

TITLE

AIR AND GASES

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P150-IA-2005	G150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-IA-2005	8 OF 8	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS